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Anchor-and-Edit:

A Controlled Evaluation of Bounded Revision for Agentic Document Ranking

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Declaration

I, Erin Ju, confirm that the work presented in my dissertation is my own. Where information has been derived from other sources, I confirm that this has been indicated in the dissertation.

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Signature

19 September 2026

Date

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Abstract

The application of agentic systems to matching research proposals with funding opportunities remains underexplored. It is unclear whether agentic reasoning and coordination can improve judgements about which funding opportunities best fit a given research proposal.

To address this gap, the dissertation first compares four agentic coordination designs for document ranking on a public benchmark under controlled conditions, with lexical and cross-encoder reference baselines (Study 1). It then evaluates the same listwise designs on a private proposal–funding matching dataset (Study 2). A domain expert provides the reference relevance judgements, while a second annotation by a non-expert applying the same rubric provides a reference for interpreting the agreement scores.

Central to the evaluation is our proposed "anchor-and-edit" architecture. Rather than reconstructing a ranking from scratch, it uses additional reasoning to select a small number of document promotions while preserving the relative order of the remaining candidates. Its aim is to correct specific ranking errors while limiting disruption to the existing result.

Anchor-and-edit improves observed ranking effectiveness over the single-pass baseline in several evaluated settings, with the strongest evidence for Biology-related queries. A controlled anchor-visibility check shows that a visible anchor did not prevent recovery of constructed Biology errors. In the applied study, although anchor-and-edit shows no clear ranking advantage for proposal–funding matching, it agrees with the expert about as closely as a second annotator applying the same rubric. With six proposals and near-ceiling ranking scores, Study 2 is reported as a feasibility check rather than an external validation. Overall, the effect of bounded revision is task-dependent. It improves observed ranking effectiveness most clearly on the Biology design domain, but shows no consistent benefit across evaluation domains, model backbones or the applied matching task. It is therefore a setting-dependent ranking strategy that requires larger-scale validation.

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Introduction

Recently, large language models (LLMs) have been organised into specialised multi-step systems, often termed *agentic* systems, that combine reasoning, tool use and revision across multiple calls (Yao et al., 2023; Madaan et al., 2023). However, it remains unclear whether coordinating these calls improves relevance judgements when the candidate documents and available evidence are fixed. This dissertation examines that question by comparing agentic coordination for document ranking and applied research proposal–funding matching. The two studies address the same underlying problem at different levels: Study 1 provides a controlled test of fixed-pool relevance ranking, while Study 2 examines whether the same anchor-and-edit principle remains useful when applied to matching research proposals with funding opportunities.

The proposed anchor-and-edit architecture addresses a specific problem in fixed-pool reranking. An initial ranking may contain useful ordering information even when its top result is wrong, whereas rebuilding the ranking from scratch may correct one error while disturbing other decisions. Anchor-and-edit therefore treats the initial ranking as an informative anchor and uses additional reasoning to make only a small number of targeted promotions. Its aim is to correct salient errors while limiting unnecessary disruption.

The evaluation is organised around two research questions:

- **RQ1:** How does agentic coordination affect fixed-pool document ranking, and how does its effect vary across components, domains and model backbones?
- **RQ2:** How does agentic coordination apply to an applied research proposal–funding matching task, and how well do its judgements align with expert relevance grades?

Study 1 compares the single-pass baseline A with three alternatives: concept-first ranking (B), alternative-interpretation coordination (C), and the proposed anchor-and-edit design (D). D is the primary design evaluated in this dissertation, while B and C provide comparison conditions for other uses of additional calls. All four designs are evaluated on a public document-ranking benchmark under controlled fixed-pool conditions, including their effectiveness, component contributions, cross-domain and cross-model variation, and call usage. Lexical and supervised neural baselines

contextualise LLM ranking performance, while comparisons against the single-pass LLM assess the additional benefit associated with coordination.

The two evaluations serve complementary purposes. The benchmark study provides a controlled test of whether agentic coordination improves ranking when the candidate pool and available evidence are fixed. This isolates the effect of the coordination design from application-specific factors. The applied study then examines whether the same strategy transfers to research proposal–funding matching, where relevance must be inferred from specialised scientific language and the full texts of proposals and funding calls, including their stated scope, eligibility criteria and other constraints. A method that performs well on a benchmark may not transfer to this setting, so the applied evaluation provides a separate test of practical use rather than a direct extension of benchmark performance. The study does not introduce a separate field-extraction stage; these factors remain embedded in the full documents supplied to the ranking systems.

Study 2 applies dataset-specific implementations of the coordination designs to a private biomedical proposal–funding dataset, where rankings are compared with expert relevance judgements and a second human annotation. A separate staged matcher that grades individual proposal–call pairs rather than ranking them is examined as an exploratory applied probe.

The dissertation contributes a bounded-revision architecture and a controlled evaluation of agentic coordination under fixed evidence. It combines component analysis, cross-setting comparisons, a call-matched control and an applied expert assessment to distinguish coordination effects from differences in task, computation and annotation.

The results show that bounded revision can improve top-1 accuracy in some settings, with the strongest evidence on Biology and less consistent effects across other domains and backbones. A controlled Biology perturbation check further shows that the visible-anchor implementation can correct recoverable ranking errors rather than simply preserving its input. In the applied study, the agentic coordination designs achieve agreement with expert judgements comparable to the non-expert rubric reference. Together, the findings support a task-dependent view of agentic coordination. One possible explanation for the uneven gains is that bounded revision helps most when the initial ranking contains recoverable errors and leaves sufficient headroom for correction, while task complexity alone does not guarantee a gain. This remains a descriptive association observed here rather than a validated rule for when to apply revision.

The remainder of the dissertation presents the background and research questions (Chapter 2), methods (Chapter 3), experiments and results (Chapter 4), and their interpretation and limitations (Chapter 5). Chapter 6 outlines future work, and Chapter 7 is the conclusion.

Background

2.1 Semantic Matching and Retrieval

Information retrieval systems rank documents by estimating which are most useful for a query. Early systems relied on word overlap and term statistics; TF–IDF and BM25 remain efficient lexical baselines (Salton and Buckley, 1988; Robertson et al., 1994). BM25 scores a document by how often it contains the query terms, weighting each term by its rarity across the collection and normalising for document length, so it rewards documents that share many distinctive terms with the query. This approach is simple and scalable, but it struggles when a relevant document expresses the same idea using different words, a problem known as vocabulary mismatch (Furnas et al., 1987). Later retrieval methods therefore use semantic representations to match conceptual relatedness as well as shared wording (Deerwester et al., 1990; Carpineto and Romano, 2012).

Neural retrieval uses learned representations to match queries and documents beyond lexical overlap. A common two-stage pipeline first retrieves candidates, then applies a more expensive reranker (Karpukhin et al., 2020; Wang et al., 2011; Lin et al., 2021). Cross-encoders jointly encode each query–document pair to predict relevance (Nogueira and Cho, 2019). Often trained on passage-ranking datasets such as MS MARCO (Nguyen et al., 2016), they provide a supervised neural reference for the ranking task studied here.

Evaluation also depends on candidate selection and label quality. Rerankers may perform less effectively when evaluation candidates differ from their training negatives (Gao et al., 2021). Incomplete relevance labels can also distort measured performance: crowdworkers sometimes prefer a neural ranker's top result to the passage labelled relevant in MS MARCO (Arabzadeh et al., 2022), while query selection can introduce survivorship bias when annotators cannot locate an answer (Gupta and MacAvaney, 2022). Measured retrieval effectiveness therefore reflects properties of the evaluation data as well as the ranking model.

Standard retrieval benchmarks evaluate whether systems can return useful passages or documents for search-style tasks (Nguyen et al., 2016; Thakur et al., 2021). For example, BRIGHT (Su et al., 2025) is a reasoning-intensive retrieval benchmark for cases in which the connection between query and document is implicit or requires

several reasoning steps, rather than surface-level semantic similarity. It is constructed so that finding the relevant documents requires reasoning about the query–document relationship rather than matching shared terms.

2.2 Document Ranking and Applied Matching

LLM-based ranking

LLMs can rerank documents by scoring individual candidates, comparing candidate pairs, or ordering a candidate set. Pointwise methods judge one query–document pair at a time, pairwise methods compare two candidates, and listwise methods return an ordering for a set of candidates (Nogueira and Cho, 2019; Zhuang et al., 2024). Pointwise scoring is easy to parallelise, whereas pairwise comparison becomes costly as the candidate set grows. Listwise ranking matches the reranking task more directly, but requires the model to compare multiple documents and produce a valid ordering.

Recent work also uses additional test-time computation to reason about relevance before producing a judgement. Rank1, for example, trains rerankers to use reasoning traces for retrieval decisions (Weller et al., 2025). Such rankings remain sensitive to candidate order, long contexts and position bias (Liu et al., 2024; Zheng et al., 2023). These limitations motivate controlled comparisons of agentic coordination.

Applied proposal–funding matching

Funding recommendation has been studied as a way to help researchers identify opportunities relevant to their work. Existing approaches often represent researchers through their publications and compare these with funding descriptions. For example, Zhu et al. (2023) formulate National Institutes of Health (NIH) grant recommendation as a classification problem, using a BERT-based model to learn relationships between publications and funding announcements. Their evaluation compares this approach with earlier retrieval baselines based on lexical matching and learned document representations.

A related approach learns from historical funding relationships. Zeng and Acuna (2024) introduce GotFunding, a learning-to-rank system trained on NIH grant–publication records. Its analysis considers textual relevance alongside features such as the temporal relationship between publications and grants. These approaches illustrate two ways to operationalise funding recommendation: predicting whether a publication and an opportunity are related, and ranking grants using evidence from previously funded research.

Evaluation in this literature combines historical records with human assessment. Zhu et al. (2023) use publication–grant links to construct reference labels and report ranking and classification metrics, including precision, recall, mean reciprocal rank (MRR) and ROC-AUC. They also conduct an external evaluation in which researchers rate recommendations tailored to their profiles. Historical links provide scalable supervision, whereas human ratings assess whether recommendations are useful to their intended recipients.

These studies demonstrate how publication histories can support funding recommendation. However, a researcher's previous publications may not fully capture the aims or requirements of a new proposal. Assessing a specific proposal directly requires judging the relevance between the proposal and call texts, which lexical overlap alone does not capture (Furnas et al., 1987). This motivates examining whether LLMs can assess relevance from proposal and call texts, and whether their judgements align with expert assessments.

Because expert relevance judgements are themselves subjective and vary between assessors (Voorhees, 2000), such an evaluation calls for a shared grading rubric and for calibration against inter-annotator agreement, rather than treating a single expert's labels as an absolute standard. Beyond the choice of scoring model, this raises a question about whether relevance assessment should be performed in a single step or organised across multiple stages.

2.3 Agentic Coordination and Evaluation

Agentic coordination

Agentic systems replace single-pass inference with multi-step execution. A language model may plan, call tools, observe intermediate results, critique its own answer, or pass state between specialised roles before producing a final output. Tool-augmented systems interleave reasoning with external actions (Yao et al., 2023; Schick et al., 2023), while self-refinement systems ask a model to critique and revise its own output over one or more rounds (Madaan et al., 2023; Shinn et al., 2023). The shared premise is that decomposing a difficult task into staged reasoning can recover errors that a single pass commits.

Agentic systems can be organised as scaffolding around one model or as coordination across multiple roles. Scaffolding changes the control flow around a single model, which may plan, answer and revise in sequence; decomposition prompting is one example

(Zhou et al., 2023). Multi-agent coordination instead separates state or instructions across stages, allowing different calls to receive different contexts or perspectives (Hong et al., 2024).

Prior work on self-refinement in agentic coordination shows that a model can use critique and a further generation step to improve an initial answer in some settings (Madaan et al., 2023; Shinn et al., 2023). Permutation self-consistency, for example, samples and aggregates several listwise orderings of the same candidates to produce a more reliable ranking, and has improved document ranking in practice (Tang et al., 2024).

On the other hand, additional stages can also introduce their own failures rather than correct errors. For instance, Huang et al. (2024) and Stechly et al. (2023) find that self-correction is unreliable and that repeated revision can degrade performance when the revising model shares the original model's blind spots. Agents instantiated from the same backbone may make correlated errors, so changing role prompts does not guarantee independent evidence (Estornell and Liu, 2024; Cemri et al., 2025). A reviewer stage can also inherit *sycophancy* (Perez et al., 2022; Sharma et al., 2023), which is the tendency of a model to agree with the answer or viewpoint it is shown rather than judge it independently.

Anchor-and-edit is the proposed coordination design in this dissertation. It is motivated by the trade-off highlighted above: additional reasoning may correct an initial error, but regenerating or reordering the entire answer may also disturb useful decisions (Madaan et al., 2023; Tang et al., 2024). To address the open question of how much freedom a revision step should have when the initial output already contains useful information, it treats the initial listwise ranking as an anchor and makes only a bounded number of targeted promotions, preserving the remaining order. The study design also aims to evaluate whether such a revision stage is affected by sycophancy.

Challenges of agentic evaluation

Evaluating agentic performance raises several challenges. The first is cost. Each stage adds model calls, tokens, latency, and provider-side variation. A coordinated architecture that receives more compute than its baseline may appear to improve because it spends more rather than because coordination itself is useful. Cost-controlled agent evaluation is therefore necessary (Kapoor et al., 2024): effectiveness must be reported alongside cost, and architecture comparisons must make the additional computation visible. A direct control can spend the same call budget on an unstructured strategy that samples and aggregates independent responses. This is the principle behind self-consistency (Wang et al., 2023) and helps separate the effect of a coordinated design from the effect

of simply issuing more calls.

Reliable evaluation also requires statistical care. Prior work has highlighted the importance of statistical reliability in information-retrieval evaluation, particularly when comparisons are based on limited query sets (Sakai, 2016). Matched-pair methods provide a complementary approach by comparing system outcomes on the same instances rather than relying only on aggregate differences (Dietterich, 1998).

2.4 Research Questions

The above literature reveals that it is unclear which organisation of multiple calls improves ranking when the candidate documents remain fixed, which components contribute, and whether gains extend across domains and backbones or reflect additional computation. In applied funding recommendation, it also remains open how well coordinated agents, in particular the bounded-revision strategy, can assess proposal–call fit. These gaps motivate two research questions.

RQ1: Fixed-pool document ranking

How does agentic coordination affect fixed-pool document ranking, and how does its effect vary across components, domains and model backbones?

Study 1 addresses this question through comparisons of complete designs, component removals and a call-matched control, with evaluation across domains and model backbones. Two hypotheses guide the analysis:

- **H1:** Anchor-and-edit is expected to achieve higher top-1 accuracy than the single-pass baseline across evaluated domains and model backbones, although the size of the effect may vary by setting.
- **H2:** The benefit of anchor-and-edit is expected to be larger when the single-pass baseline has weaker ranking accuracy, because more headroom remains for targeted revision.

RQ2: Applied proposal–funding matching

How does agentic coordination apply to an applied research proposal–funding matching task, and how well do its judgements align with expert relevance grades?

Study 2 addresses this question descriptively on a small applied dataset. It examines agreement between model grades and expert judgements, using a non-expert annotation under the same rubric as a reference, and separately assesses changes in ranking.

Methods

This chapter first outlines the study structure, then introduces the two datasets and their analytical roles, defines the ranking and grading tasks, defines the evaluation metrics and statistical analysis, and finally describes the model architectures and coordination methods.

3.1 Study Structure

Study 1 addresses RQ1 through fixed-pool document reranking, where each system ranks the same candidate documents for a given query. It compares coordination designs and examines component contributions, variation across domains and model backbones, and the role of additional model calls.

Study 2 addresses RQ2 through research proposal–funding call matching. Dataset-specific implementations of the A–D coordination designs rank the six funding calls for each proposal, and their rankings are assessed against expert relevance grades. The applied task is evaluated both on the main backbone and across a cross-model panel of proprietary and open-weight backbones, as in Study 1's cross-model analysis, so that the applied findings are not tied to a single model. A decomposed staged matcher that grades individual pairs is examined as an exploratory probe in appendix A.4.

Study 1 provides the primary inferential analysis, while Study 2 provides a descriptive assessment against expert judgements. Figure 3.1 summarises the datasets, tasks and outputs.

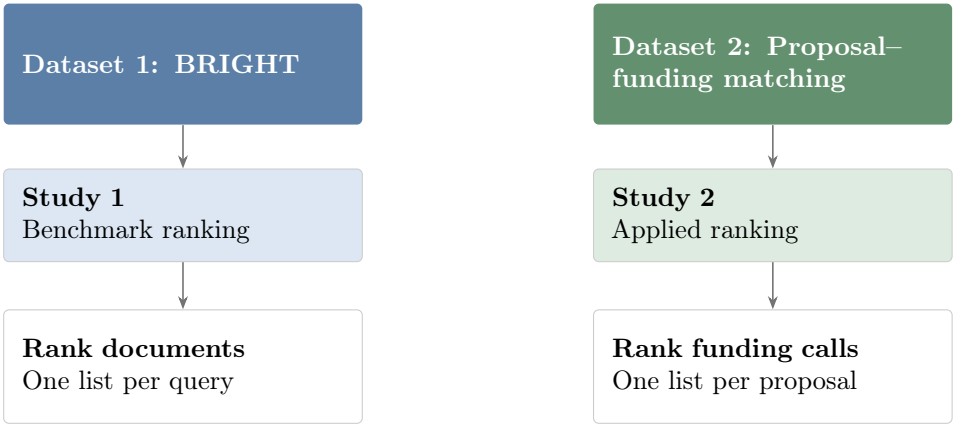


Figure 3.1: Study structure. Study 1 uses the BRIGHT document-ranking benchmark for the primary inferential comparison; Study 2 applies dataset-specific implementations of the A–D designs on a small, single-expert biomedical proposal–funding matching dataset as a descriptive external test.

3.2 Datasets

3.2.1 Study 1: BRIGHT Benchmark

Original benchmark

BRIGHT (Su et al., 2025)[1] is a retrieval benchmark for queries whose relevant documents often cannot be identified through surface matching alone. It contains naturally occurring queries from twelve domains built from several sources, seven of them drawn from StackExchange communities; table 3.1 summarises the benchmark as used here.

The present experiment uses four of BRIGHT's StackExchange[2] domains (table 3.2). In these domains each query is a user's StackExchange post, and a document is treated as relevant if it is a web page cited in the post's answers; in the original benchmark these labels are validated by multiple annotators, and only documents that all annotators agree support the query are retained (Su et al., 2025). The corpus for each domain is the collection of such cited web documents, which lets candidate pools be constructed consistently across the four domains. These four domains were selected because they share a common StackExchange data structure, which permits consistent

[1]The BRIGHT benchmark is publicly available at `https://huggingface.co/datasets/xlangai/BRIGHT` and `https://github.com/xlang-ai/BRIGHT`.

[2]StackExchange is a network of topic-specific question-and-answer communities; BRIGHT draws these domains from four such communities: `https://stackexchange.com`.

pool construction, while spanning distinct scientific and applied subject areas that give a varied transfer set. The prompts and coordination designs were developed on Biology; Earth science, Sustainable living, and Psychology were then evaluated without further design changes.

Candidate-pool construction

From this source, a fixed pool of 100 candidate documents is constructed for each query. A 100-document pool provides a demanding reranking setting while keeping the rendered candidate set within the model context window. All of the query's relevant documents are inserted, and the remaining slots are filled with hard negatives: the highest-scoring corpus documents under a TF–IDF retriever (Salton and Buckley, 1988) that are neither relevant nor marked as excluded by BRIGHT. Hard negatives are used instead of random negatives so that the distractors share substantial lexical content with the query, making the ranking task non-trivial. Documents are truncated to their first 600 characters to bound and equalise input length across candidates; they are presented under opaque identifiers so that no system can exploit semantic cues from document metadata; and the pool order is shuffled with a fixed seed so that presentation order is held constant across the compared systems rather than confounding their differences. Relevant documents are therefore present by construction, so the experiment evaluates fixed-pool reranking rather than end-to-end retrieval.

An eligibility criterion retains only queries with between one and six relevant documents present in the pool. The lower bound ensures that at least one relevant document is available to rank. The upper bound limits variation in the number of relevant documents per query rather than query difficulty itself, since a pool containing many relevant documents makes a top-ranked hit more likely. The criterion is applied before variant evaluation and identically across arms, so it changes the evaluated query set but does not favour any coordination variant within that set.

Table 3.1: Characteristics of the BRIGHT benchmark.

Property	**Value**
Candidate set	relevant documents and lexically similar hard negatives
Set size	100 per query
Queries	342 retained across four domains
Relevant per query	one to six
Relevance	binary
Labels	web pages cited in the post's answers, multi-annotator verified

Table 3.2: BRIGHT domains used, with query counts before and after the eligibility criterion.

Domain	**Queries**	**Retained**	**Role**
Biology	103	97	design + evaluation
Earth science	116	90	evaluation
Sustainable living	108	80	evaluation
Psychology	101	75	evaluation
Total	428	342	

3.2.2 Study 2: Applied Proposal–Funding Matching

Study 2 is an applied case study of biomedical research proposal–funding matching, which provides an external test of the coordination designs but is treated as a descriptive evaluation rather than a powered external validation. The dataset comes from live matching work rather than being constructed for this study. Six research proposals are each evaluated against the same six funding calls, producing 36 proposal–call pairs, and a practitioner responsible for the matching task assigns each pair one of three relevance grades.

Unlike BRIGHT, the six calls constitute the complete candidate set considered for each proposal. The dataset is proprietary and is not released with this dissertation; the underlying texts, relevance grades, derived rankings and analysis artefacts are retained internally. Table 3.3 summarises the dataset.

Table 3.3: Characteristics of the applied proposal–funding matching dataset.

Property	**Value**
Candidate set	all six funding calls considered for each proposal
Queries	6 proposals, giving 36 proposal–call pairs
Relevant per query	zero to three
Relevance	three-level
Document input	full proposal and call text, no length cap or truncation
Labels	practitioner responsible for the matching task

Some proposals contain no acceptable candidate. The ranking architecture always returns a permutation and does not implement abstention. Ranking metrics concern proposals with at least one acceptable candidate; agreement with expert grades is evaluated over all 36 proposal–call pairs.

Each proposal and funding call is supplied to the systems as its complete document text, with no length cap or truncation (the longest proposal is about 28,000 characters

and the longest call about 41,000). No field extraction or chunking is applied, so the systems must locate the relevant constraints, such as deadlines and eligibility criteria, within the full text. This choice keeps the evaluation focused on whether agentic coordination can improve ranking from the available evidence, rather than introducing a separate preprocessing component whose effect would be difficult to distinguish from the coordination design.

Each proposal–call pair is judged on a three-level relevance scale, defined by a fixed written rubric developed in collaboration with the domain expert who provides the gold labels, so that the codified criteria reflect the practitioner's actual grading practice rather than being imposed after the fact:

- `match`: the proposal clearly falls within the call's stated therapeutic areas, scope or eligibility;
- `borderline`: a partial fit that overlaps on some dimension (disease area, modality or mechanism) but has a scope, eligibility or sponsor-specific mismatch that leaves suitability genuinely uncertain;
- `no_match`: outside the call's stated scope, or in violation of a stated constraint (wrong disease area, wrong modality, an explicit exclusion, or a sponsor or institutional conflict).

The three-level scale was selected to distinguish clear fit, uncertain fit and no fit, consistent with the expert rubric. It retains an intermediate category for ambiguous cases without requiring annotators to distinguish additional degrees of suitability.

The domain expert responsible for the proposal–funding matching task assigned all 36 gold labels from the proposal and call texts. The rubric prioritises hard-constraint violations, then clear fit, with remaining cases graded `borderline`. It is reproduced in appendix A.3 and supplied verbatim to the matching systems.

The author independently re-graded all 36 pairs using the same rubric without viewing system outputs. The author is not a domain expert, so this second annotation measures how far the written rubric transfers the grading task to a non-expert, and is referred to below as the *non-expert rubric reference*. It is a reference point for rubric transferability, not an estimate of human ceiling performance or of intrinsic task ambiguity. The expert grades remain the sole evaluation reference.

3.3 Task Definition

Study 1 and Study 2 share a fixed-pool listwise ranking task. For each query q, a candidate set D_q containing N items is constructed once, persisted, and held constant across the systems being compared. Each system receives the complete rendered candidate set and returns a permutation of its identifiers. Systems cannot retrieve additional documents or expand D_q. The comparison therefore concerns how they process the available evidence rather than which documents they retrieve.

In Study 1, q is a benchmark query and D_q contains 100 documents. In Study 2, q is a research proposal and D_q contains the six funding calls. The resulting list orders candidates by relevance to the query.

3.4 Reference Baselines

To contextualise LLM ranking performance, two conventional rerankers are evaluated on the same fixed candidate pools. BM25 is a lexical scoring function based on query–document term overlap (Robertson et al., 1994); it tests how far surface matching alone can solve the task. The MS MARCO-trained cross-encoder jointly encodes each query–document pair to predict relevance (Nogueira and Cho, 2019; Nguyen et al., 2016), providing a supervised neural reference. Together, these baselines show whether the task requires more than standard lexical or neural reranking.

3.5 Evaluation Metrics and Statistical Analysis

The metric set follows the different roles of the two studies. Study 1 tests ranking effectiveness against benchmark relevance labels, whereas Study 2 asks how closely model judgements agree with expert assessments in an applied setting. Statistical tests and uncertainty intervals are used for the primary Study 1 comparisons; Study 2 results are descriptive because they contain only six proposals and one expert reference.

3.5.1 Study 1: Ranking Effectiveness

Study 1 uses three ranking metrics, computed over the evaluated query set Q. For a query q, the system returns a ranking of the candidate documents; let r_q be the position of the first relevant document and rel_i the relevance of the document at rank i (binary

in Study 1).

- **Hit@1** (top-1 accuracy) is the fraction of queries whose top-ranked document is relevant, the top-1 case of the top-k accuracy used in passage retrieval (Karpukhin et al., 2020). It is the primary measure for Study 1 because BRIGHT relevance is binary and the question is whether the best recommendation is correct. Formally,

$$\text{Hit@1} = \frac{1}{|Q|} \sum_{q \in Q} \text{rel}_1^{(q)}.$$

- **MRR** (mean reciprocal rank) averages the reciprocal rank of the first relevant document, giving a less brittle view of near-misses (Voorhees, 1999). Formally,

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{r_q}.$$

- **nDCG** (normalised discounted cumulative gain) measures the graded quality of the whole ordering while discounting lower positions (Järvelin and Kekäläinen, 2002). In Study 1 it is secondary because full rankings are not available in every saved export. With a rank cutoff k,

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{\text{rel}_i}{\log_2(i+1)}, \qquad \text{nDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k},$$

  where IDCG@k is the DCG@k of the ideal ordering, so nDCG@$k \in [0, 1]$.

3.5.2 Study 2: Applied Agreement and Ranking

Agreement with the expert is measured over all 36 proposal–call pairs using three-level quadratic-weighted Cohen's κ (Cohen, 1968). The grades are `no_match`, `borderline` and `match`; quadratic weighting is used because the grades are ordinal, so a disagreement is penalised more heavily than an adjacent one, which receives partial credit. On this scale $\kappa = 1$ is perfect agreement and $\kappa = 0$ is the agreement expected by chance under the metric, so larger values indicate closer agreement after accounting for chance. This agreement is reported for each of A, B, C and D; because anchor-and-edit is a bounded revision of the ordering rather than a re-grading step, it inherits A's grades and changes only their order, so D's agreement is identical to A's by construction, and the κ comparison is informative only among A, B and C. All applied metrics use a single presentation-order seed and are interpreted descriptively.

Whether anchor-and-edit improves the applied ranking is assessed separately, through

ranking quality. This is measured primarily with nDCG@6, which evaluates the graded ordering of all six candidate calls using linear relevance gains (`match`= 2, `borderline`= 1, `no_match`= 0). Linear gains reflect the ordinal three-level rubric without assuming that a `match` is more than twice as valuable as a `borderline` call. A proposal contributes to nDCG@6 when it has a non-zero ideal ranking (at least one `match` or `borderline`); a proposal graded `no_match` on every call has an undefined ideal ranking and is excluded. Hit@1 and MRR are supplementary match-only measures, computed over the proposals with at least one `match`. Because few proposals meet these conditions on this small dataset, these top-focused metrics are coarse and are not used for substantive conclusions. All 36 proposal–call pairs remain in the agreement analysis.

3.5.3 Statistical Analysis

Hit@1 contrasts report fix and break counts and use two-sided exact McNemar tests on discordant query outcomes (McNemar, 1947). The unadjusted significance threshold is $p < 0.05$. Five replications of the same queries are analysed separately and are not treated as independent query samples. The cross-domain family of 12 comparisons uses the Bonferroni correction (Dunn, 1961) as a conservative sensitivity check across that larger family; the three component-removal contrasts use the uniformly more powerful Holm correction (Holm, 1979) within each replication. These paired tests are appropriate because every system is evaluated on the same queries and candidate pools, while fix and break counts show whether an aggregate difference comes from correcting baseline errors or disrupting baseline successes.

Paired Hit@1 and MRR differences use pointwise 95% query-cluster bootstrap intervals (Efron, 1979). Queries are resampled while retaining their repeated observations together. The primary and follow-up Biology series are reported separately and are not pooled. Standard deviations across replications describe execution variation, not confidence intervals over queries. The query is the resampling unit because repeated executions of one query are not independent observations about the query population.

3.6 Model Architecture and Coordination Methods

3.6.1 Architecture

All variants are implemented as explicit Python pipelines that carry the query, candidate set, current ordering and any intermediate representations generated by the active variant. The pipeline topology is shared across datasets, while prompts and candidate

rendering are dataset-specific.

A deterministic rendering function maps each candidate to a stable identifier, and model responses are parsed as permutations of those identifiers. Truncated, duplicated or incomplete outputs are repaired by retaining the valid parsed prefix and appending missing candidates in their original order, a rule chosen because it preserves every valid ordering decision the model did make while applying the smallest deterministic change needed to produce a complete permutation. Repairs and call failures are logged separately from effectiveness, and differences in the information retained by each export constrain which analyses can be reconstructed, as discussed in section 4.1.2.

3.6.2 Listwise Coordination Variants

Four listwise variants are evaluated on both datasets. Variant A is the single-pass reference condition, while B–D implement coordination mechanisms motivated by section 2.3. They differ in how freely additional calls can alter the final ordering. This matters because ranking errors may be local or global: a method may need to promote one missed relevant document without discarding a useful overall ordering. Anchor-and-edit addresses this problem by separating error diagnosis from a bounded edit operation. Comparing D with B and C, rather than with A alone, helps assess whether any advantage is specific to bounded revision rather than to coordination or additional calls in general. Figure 3.2 summarises their control flow. The exact prompt used by each variant is reproduced in appendix A.2 for Study 1 and appendix A.3 for Study 2.

A: Single-pass baseline. Variant A consists of one model call using the A-prompt over the complete rendered candidate set. The resulting permutation is the final ranking. It contains no explicit coordination and serves as the reference condition for paired comparisons.

B: Concept-guided reranking. Variant B first abstracts the information need into a higher-level semantic representation and then ranks the candidates conditional on that representation. It tests whether explicit concept abstraction improves ranking when relevance depends on conceptual rather than surface correspondence. B requires two model calls per query.

C: Alternative-interpretation coordination. Variant C generates two alternative interpretations of the query–document relevance relationship: a *surface-oriented* reading,

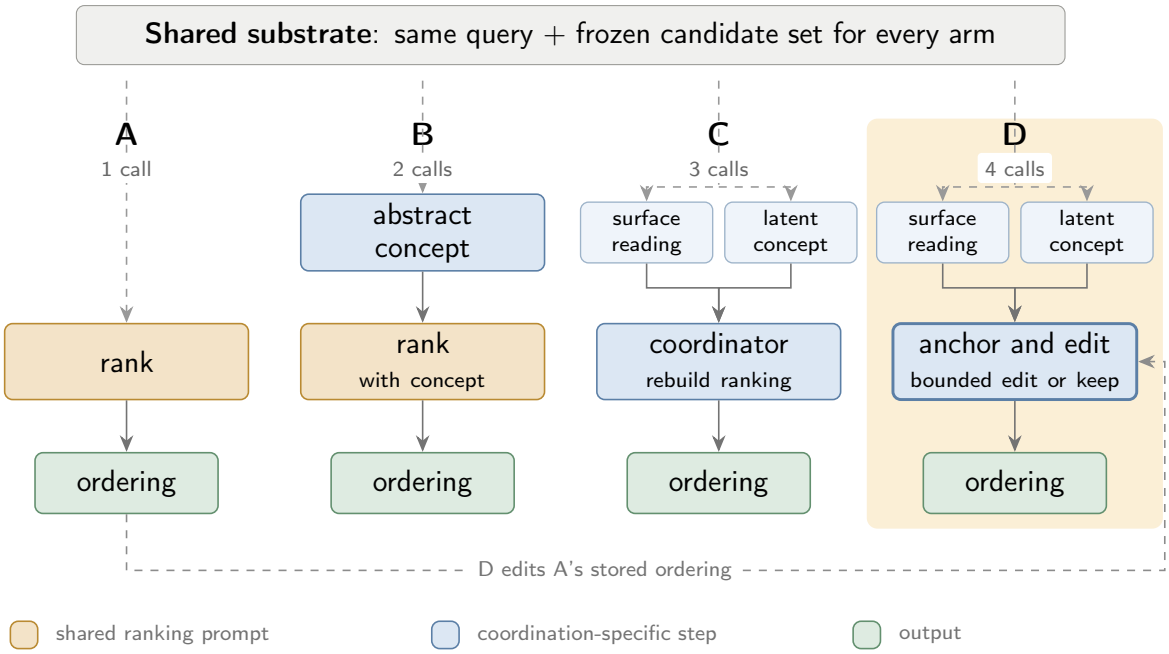


Figure 3.2: Listwise coordination designs A–D; the shaded column marks the proposed design, anchor-and-edit (D). A and B rank with the shared base ranking prompt (the A-prompt), and variant B additionally conditions the ranking on the abstract concept; C and D instead use a coordination-specific step. D uses three additional calls after A, or four calls in total.

which captures the literal terms, answer type and explicit constraints an answering document would contain, and a *latent-concept* reading, which names the underlying mechanism, principle or concept that actually answers the query even when it is not stated on the surface. A coordinator then integrates these interpretations into a new ranking. C tests whether preserving alternative interpretations before coordination can recover cases in which early commitment is detrimental. It requires three model calls per query.

D: Anchor and edit. Variant D treats A's ranking as an anchor and uses the surface-oriented and latent-concept interpretations from C to select targeted promotions. Revision is bounded to at most two promotions in Study 1 and one in Study 2. These caps are fixed design parameters, specified in the prompt and enforced after parsing. D therefore revises the existing ranking rather than rebuilding it, using four calls in total: one for A and three additional calls for D.

The coordinator receives the anchor differently in the two studies. In Study 1 the anchor is hidden so that the coordinator identifies promotions independently, and these are applied to A's stored ranking in code; in Study 2 the six-item anchor is shown because the applied coordinator explicitly decides whether the complete ranking should

be preserved or edited. The designs therefore share the anchor-and-edit principle but differ in anchor visibility, which may affect cross-study comparisons.

Experiments and Results

This chapter presents the experiments and results in the order of the research questions. It first defines the controls shared across the studies, including candidate-pool parity, model and decoding settings, replication and output validity. It then reports the primary Study 1 comparison of the four coordination designs and its outcome and case analysis, followed by cross-model, cross-domain, component and call-matched analyses. Study 2 is presented next as an applied evaluation, followed by an anchor-visibility perturbation check and then computational cost and outcome stability. Each experiment gives its purpose and specific setup immediately before its results; common evaluation metrics and statistical procedures are defined in section 3.5.

4.1 Common Experimental Settings

4.1.1 Shared Controls and Execution

Within each dataset's A–D listwise comparison, experimental parity is enforced in three ways. First, all variants receive the same candidate set, in the same initial order, through the same deterministic rendering function. Second, all variants are derived from a frozen dataset-specific base relevance prompt, hereafter the A-prompt. Coordination variants may add intermediate representations or mechanism-specific instructions, but the candidate representation and relevance objective are not independently retuned across arms; the exact prompt for each variant is reproduced in appendices A.2 and A.3. Third, within each run, all variants use the same model backbone and decoding configuration. Model-specific deviations from this parity are reported with the corresponding experiment.

Computational expenditure is not equalised across variants because coordination introduces additional model calls and token consumption. Cost is therefore treated as an outcome and reported alongside ranking effectiveness.

Main backbone

The main backbone is the dated Haiku 4.5 snapshot listed in table 4.4. It was chosen as the main backbone because it is a small, low-cost model in the Claude family with strong

listwise instruction-following, and because it is available as a pinned dated snapshot that fixes the exact model version for reproducibility. This balance of instruction-following, context capacity and inference cost made it feasible to run the full experimental suite—multiple designs, domains and replications—at scale. It is shared across the four-domain A–D comparison, component experiments and applied evaluation. Additional backbones are used for the Biology A/D panel. Each experiment states its own sample and replication count below.

Decoding

Where the sampling temperature can be set, it is set to zero, to reduce avoidable sampling variation between otherwise identical conditions. The four reasoning-capable open-weight models served through Together AI[1] run with `enable_thinking=false`. Some providers do not expose identical decoding controls; any such deviations are stated with the corresponding model rather than assumed away. Where a reasoning-effort control is available, it is set to its lowest setting so that inference-time reasoning is not introduced as an uncontrolled second source of coordination.

Replication

Temperature zero does not guarantee bit-identical outputs. Repeated executions assess variation within a configuration; replication counts are given with each experiment. The appendix's staged matcher is also repeated ten times.

Seeds recorded for each run control the order in which candidates are rendered into the prompt. They are not passed to the provider, so a run is not reproducible from its seed alone. Replicate variation therefore combines input-order sensitivity with provider-side variation. It is not sampling uncertainty over queries, and the two are not reported in a single interval.

4.1.2 Output Validity and Reproducibility

The parser repairs incomplete rankings by appending missing identifiers to the tail, while call failures are recorded separately. A high repair rate limits the evidential status of the result because the measured output then partly reflects the repair heuristic rather than the model.

[1]Together AI is a commercial cloud platform that hosts open-weight models behind a shared inference API: `https://www.together.ai`.

Comparisons use complete-case masks appropriate to their design. The shared cross-domain A–D analysis contains 97 Biology queries in replications 1–4 and 96 in replication 5 because of one C-arm failure; the other domains retain all queries. Cross-model A/D and component-follow-up contrasts use their own matched valid samples, so failures in unrelated arms do not remove a query. Failure-policy sensitivity results retain incoming orderings and are reported separately.

Saved-output provenance

Reported results are reconstructed from saved outputs, whose completeness varies by experiment. The primary BRIGHT exports retain the resolved model and per-query outcomes but not full rankings, top-ranked identifiers or execution timestamps; primary per-domain nDCG and some run provenance therefore cannot be recovered. The separate five-replication component follow-up retains full rankings, raw responses, costs and timestamps, enabling the direct component tests reported here.

Stability analyses consequently describe agreement among saved outputs rather than independently verified reruns. Recorded timestamps distinguish some applied runs but do not establish independent execution of every stage or replication. Provider-side execution and presentation order remain unisolated sources of variation.[2]

4.2 Study 1: Document Ranking

The experiments first establish conventional non-LLM reranker baselines, then compare A–D on the design domain, and examine transfer across backbones and domains, component contributions, and an alternative use of additional calls. Unless labelled as adjusted, p-values are uncorrected and tests are conducted within each replication.

4.2.1 Reference Baselines

Setup

On the Biology pools (100 candidates, 97 complete-case queries), BM25 uses the standard Okapi parameters $k_1 = 1.5$, which controls how quickly repeated query-term matches saturate, and $b = 0.75$, which controls document-length normalisation, over

[2]The code used to run the experiments is available at `https://github.com/erinju7/agentic-ranking-bounded-revision`; the proprietary Study 2 dataset and expert labels are not released.

whitespace-tokenised text. These are the standard default Okapi settings. The cross-encoder `ms-marco-MiniLM-L-6-v2` (Reimers and Gurevych, 2019; Wang et al., 2020) scores each query–candidate pair, with queries truncated to 200 tokens and a maximum sequence length of 512, following the checkpoint's input limit and leaving budget for the candidate text. Both methods are deterministic and are run once. Random ranking has an expected Hit@1 of 0.032 (the mean ratio of relevant documents to pool size).

Results

Table 4.1: Reference baseline results on BRIGHT Biology (97 queries, 100 candidates per query).

Reranker	**Hit@1**	**MRR**
Lexical (BM25)	0.052	0.156
Cross-encoder (`ms-marco-MiniLM-L-6-v2`)	0.072	0.155

BM25 reaches Hit@1 of 0.052 and the cross-encoder 0.072, versus 0.032 under random ranking; their MRR values are 0.156 and 0.155, respectively. Both methods place a relevant document first only about 5–7% of the time, barely above the random floor of 3.2%, and their low MRR (near 0.15) indicates weak overall ranking quality. These two baselines therefore perform poorly on the pools: surface term overlap and a standard supervised cross-encoder are not sufficient here, consistent with BRIGHT being designed so that relevance often depends on reasoning beyond surface similarity rather than shared query terms. These baselines therefore establish a low reference point against which the LLM coordination designs, which reach Hit@1 above 0.5 (table 4.2), are compared.

4.2.2 Comparing Coordination Designs

Purpose and setup

This experiment tests H1: whether explicit coordination improves ranking relative to a single pass. A, B, C and D rank the same pools of 100 candidates on Biology using the pinned main backbone `claude-haiku-4-5-20251001` at temperature zero. Five replications retain 97 shared complete cases in replications 1–4 and 96 in replication 5. Biology was used to develop the designs; these results therefore establish the starting comparison, not evidence of transfer.

Results

All four LLM designs achieve higher observed Hit@1 than BM25 (0.052) and the cross-encoder (0.072) (table 4.1), with scores ranging from 0.558 to 0.680 (table 4.2). Even the single-pass baseline A therefore provides a substantial improvement over these reference baselines on the evaluated Biology pools.

Among the coordination designs, D achieves the highest mean Hit@1 and improves on A in all five replications at the unadjusted significance threshold ($p = 0.012$–0.041). The paired D–A difference is +0.113 (pointwise 95% CI $[+0.029, +0.198]$); in practical terms, D raises the proportion of queries with a relevant document ranked first from 56.6% to 68.0%, an improvement of 11.3 percentage points. Neither B nor C shows a statistically significant improvement over A in any replication.

Table 4.2: Mean Hit@1 for the four coordination designs on BRIGHT Biology over five replications (97 shared complete cases in replications 1–4; 96 in replication 5). The final column counts the replications in which the design improves on the single-pass baseline A at the unadjusted McNemar threshold.

Design	**Hit@1**	**Significant improvement over A (of 5)**
A: single-pass	0.566	—
B: concept-first	0.614	0/5
C: alternative interpretation	0.558	0/5
D: anchor-and-edit	**0.680**	**5/5**

Ranking transitions

The following analysis combines pooled transition counts across five Biology replications with a taxonomy and illustrative cases from one capture run. The categories describe what changed, not why the model made the decision. Because relevant documents are included in the pool, an unrecovered deep-ranked document is a ranking miss rather than a retrieval failure.

Across five pooled Biology replications, D edits 96% of rankings and retains the original ordering in the remaining 4%. It produces 77 corrections of an incorrect top rank and 22 adverse changes to a correct top rank, or about 3.5 corrections per adverse change. The net effect on Hit@1 is therefore positive, with limited damage from adverse changes.

Table 4.3: Ranking outcomes from one Biology replication ($n = 97$), grouped by whether A's top-ranked document was already relevant.

Category	n	%	What happened
A's top-1 wrong (42 queries): D can rescue or miss			
Near-miss rescue	10	10.3	A ranked a relevant document 2–5; D moved it to rank one
Deep recovery	5	5.2	A ranked a relevant document 6 or lower; D moved it to rank one
Uncorrected near-miss	11	11.3	A ranked a relevant document 2–5, but D did not move it to rank one
Mid-ranking miss	6	6.2	A ranked a relevant document 6–10, and D did not move it to rank one
Deep ranking miss	10	10.3	A ranked a relevant document below rank 10, and D left it below rank one
A's top-1 correct (55 queries): D can preserve or break			
Kept-correct	49	50.5	A ranked a relevant document first, and D preserved it
Promotion error	6	6.2	A ranked a relevant document first, but D moved a non-relevant document above it

The taxonomy shows a positive but asymmetric pattern. D rescues some of A's incorrect top ranks while usually preserving rankings that are already correct. All remaining misses are ranking misses rather than retrieval failures, because every relevant document was already in the pool.

Qualitative case analysis

Four cases from the same capture illustrate the main outcome categories (Figure 4.1): two of successful revision and two of its failure modes. They are illustrative single-run observations, not additional quantitative evidence.

Near-miss rescue ($n = 10$)

Query. "why does an electrical shock freeze up muscles?"

A. Ranks first a document that *describes* the symptom; the relevant document, which *defines* tetanic contraction, sits at rank three.

D. Infers the latent concept (tetanic contraction) and promotes the definition to rank one ($3 \rightarrow 1$).

Deep recovery ($n = 5$)

Query. "Do large animals often evolve into smaller animals?"

A. Ranks surface documents on specific extinctions first; the relevant document, which *defines* Cope's rule and its reversal, sits at rank thirteen.

D. Infers the latent concept (directional body-size evolution) and promotes the definition to rank one ($13 \rightarrow 1$).

Promotion error ($n = 6$)

Query. "are there any documented cases of extinction of a species of fungus? I imagine it would be hard to detect something like this."

A. Correctly ranks *Prototaxites*, an extinct fungal genus, first.

D. Over-reads the aside about detection difficulty and promotes an IUCN extinction-criteria document above the relevant one ($1 \rightarrow 3$).

Uncorrected near-miss ($n = 11$)

Query. "why do some vegetables taste bitter? Which chemical causes bitterness?"

A. Ranks first a document on glucosinolates; the relevant document, also on glucosinolates but in pungent plants, sits at rank four.

D. Infers the right concept but promotes a sibling document, leaving the near-miss uncorrected ($4 \rightarrow 4$).

Figure 4.1: Four illustrative cases from one Biology capture: two rescues (a near-miss and a deep recovery) and two failures (a promotion error and an uncorrected near-miss).

The electrical-shock example illustrates the intended opportunity. A ranks a document describing the symptom first, while the relevant explanation of tetanic contraction is third. D's latent-concept interpretation identifies the explanatory concept and promotes that document to first place, and the remaining candidates retain their relative order. This is a concrete example of the intended design: improving the leading recommendation through a targeted promotion while preserving the rest of the initial ordering. The deep-recovery case shows the same mechanism operating from a much lower starting rank, whereas the promotion error and the uncorrected near-miss show its two failure modes: overgeneralising a question, and choosing a plausible neighbouring document when several candidates share the same concept.

This pattern suggests that recoverable ranking errors may help explain the Biology gains: the baseline leaves room for improvement, and some relevant documents are already close to the top. However, revision also introduces six top-1 errors in the captured run, and the component ablations do not establish which input causes the benefit; Biology

Table 4.4: Reported model backbones and decoding settings. All backbones decode at temperature 0; the four reasoning-capable open-weight models additionally run with reasoning ("thinking") disabled for parity with the direct-answer closed models.[a]

Model	Family	Weights	Decoding	k
`haiku-4.5`	Claude	closed	temp 0	5
`claude-opus-4-6`	Claude	closed	temp 0	2
`claude-sonnet-4-6`	Claude	closed	temp 0	2
`gemini-3.5-flash`	Gemini	closed	temp 0	2
`gemini-3.7-flash`	Gemini	closed	temp 0	2
`meta-llama/Llama-3.3-70B-Instruct-Turbo`	Llama	open	temp 0	2
`deepseek-ai/DeepSeek-V4-Flash-0731`	DeepSeek	open	temp 0; thinking off	2
`openai/gpt-oss-20b`	GPT-OSS	open	temp 0; thinking off	2
`openai/gpt-oss-120b`	GPT-OSS	open	temp 0; thinking off	2
`Qwen/Qwen3.5-9B`	Qwen	open	temp 0; thinking off	2

[a] Llama-3.3-70B has no reasoning mode and runs direct-answer natively, so no thinking flag was set; the flag (`enable_thinking=false`) was applied to Qwen, both GPT-OSS models, and DeepSeek. Reasoning left enabled otherwise exhausted the token budget on hidden chain-of-thought and truncated the ranking.

was also the design domain. The evidence therefore supports bounded revision as a useful design in this setting, while leaving its mechanism and transfer conditions open.

4.2.3 Cross-Model Evaluation

Purpose and setup

To examine H2 and the transfer of the D–A effect across backbones, the cross-model comparison is restricted to the A/D contrast on BRIGHT Biology with a candidate pool of 100. The panel is restricted to A and D to test whether the effect of the proposed anchor-and-edit architecture recurs across backbones relative to the single-pass baseline. B and C are retained for the primary controlled comparison rather than repeated across all backbones. The panel is therefore not a full repetition of the A–D study for every backbone; task, prompt family, candidate pool and decoding settings are held as close as provider interfaces allow.

The main panel contains ten tested backbones with stated model identifiers and decoding configurations (table 4.4). Coverage repair and call failure rates are reported as diagnostics.

The cross-model panel contains ten reported models. It includes both closed-weight and open-weight backbones so that the analysis is not limited to a single provider family. The k column reports the number of replications for the corresponding experimental role.

Panel overview

Table 4.5 reports the single-pass baseline against anchor-and-edit on BRIGHT Biology for the ten reported backbones, ordered by single-pass strength. The replication count differs by backbone by design: the primary backbone is reported at $k = 5$ to characterise execution variability more reliably for the main inferential comparison, while the panel backbones are reported at $k = 2$ because the panel is a transfer check across models rather than the primary analysis. The model roster is specified in section 4.2.3, and release dates for the reported backbones are listed in appendix A.1.

Table 4.5: A/D comparison on BRIGHT Biology (pool 100). A and D are the mean Hit@1 of single-pass (A) and anchor-and-edit (D), D−A is their difference, and k is the number of replications. D sig. counts the replications (out of k) in which D improves on A at the unadjusted McNemar threshold ($p < 0.05$). Repair rate is the proportion of outputs with incomplete candidate lists, which are deterministically completed by appending missing IDs at the end. Call failure rate is the proportion of queries with no usable response; these queries are excluded from paired comparisons.

Model	k	**A**	**D**	**D−A**	**Repair rate**	**Call failure rate**	**D sig.**
`Llama-3.3-70B`	2	0.356	0.505	**+0.149**	48 %	0 %	**2/2**
`gpt-oss-20B`	2	0.460	0.572	**+0.112**	31 %	4 %	**2/2**
`Qwen3.5-9B`	2	0.482	0.580	+0.098	94 %	1 %	0/2
`gpt-oss-120B`	2	0.526	0.644	**+0.118**	39 %	0 %	**2/2**
`haiku-4.5`*	5	0.567	0.680	**+0.113**	75 %	0 %	**5/5**
`opus-4-6`	2	0.572	0.613	+0.041	10 %	0 %	0/2
`gemini-3.5-flash`	2	0.582	0.598	+0.015	11 %	0 %	0/2
`DeepSeek-V4-Flash`	2	0.598	0.624	+0.026	35 %	0 %	0/2
`gemini-3.7-flash`	2	0.608	0.608	+0.000	47 %	0 %	0/2
`sonnet-4-6`	2	0.644	0.665	+0.021	19 %	0 %	0/2

* Main backbone. Per-replicate McNemar p-values are: Llama-3.3-70B, 0.003 and 0.003; gpt-oss-20B, 0.027 and 0.041; gpt-oss-120B, 0.019 and 0.029; Qwen3.5-9B, 0.078 and 0.052; all higher-baseline rows, 0.22 or larger. The Haiku panel uses all 97 queries in every replication, unlike the shared four-arm sample in table 4.6.

Ranking accuracy comparison

Figure 4.2 shows substantial variation in the D–A gain across backbones. The largest gains occur for models with weaker single-pass baselines, whereas models with stronger A performance show little or no improvement. Among the ten reported backbones, Llama-3.3-70B, gpt-oss-20B and gpt-oss-120B reach the unadjusted threshold in both replications, and Haiku does so in all five. Qwen's gains do not reach that threshold. The five models with A at or above 0.57 have observed gains from 0.000 to 0.041, with no significant replication. Some backbones have high repair rates in table 4.5 (75 % for the primary Haiku backbone, 94 % for Qwen), but repair only appends missing identifiers to the tail of a ranking and does not change the top-ranked item, so it does not affect the Hit@1 comparison; the primary backbone's top-1 output is complete in every case.

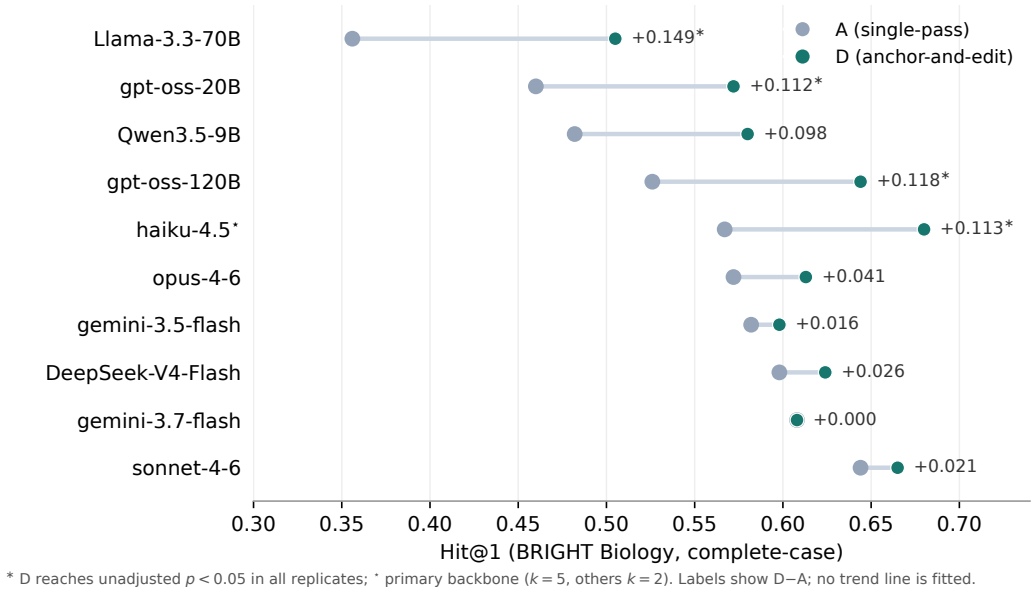


Figure 4.2: A and D Hit@1 by backbone on Biology; labels show D–A differences.

The figure also permits four exploratory comparisons of model attributes, each involving only a few models and reported descriptively (release dates are listed in Appendix A.1).

- **Release generation (Gemini).** Gemini-3.7-Flash has a higher baseline than Gemini-3.5-Flash (0.608 versus 0.582) but no larger D–A gain (+0.000 versus +0.015): the newer model ranks better in a single pass yet gains nothing from revision.
- **Model tier (Claude).** Opus-4-6, the larger Claude tier, has a *lower* single-pass baseline than Sonnet-4-6 (0.572 versus 0.644) yet a larger gain (+0.041 versus

+0.021). Together with the Gemini pair, these examples show that the observed gains differ across model families, although each comparison contains only two models.

- **Model size (gpt-oss).** The gpt-oss pair gives a limited size comparison. gpt-oss-120B has a higher baseline than gpt-oss-20B (0.526 versus 0.460) with a near-identical gain (+0.118 versus +0.112): larger size tracks a stronger baseline here, not a larger revision benefit.

- **Open versus closed weights.** Open-weight models gain more on average, but this is confounded with baseline strength and family: DeepSeek-V4-Flash gains only +0.026 despite being open-weight, whereas the closed-weight Haiku gains +0.113.

Overall, the observed variation is descriptively consistent with the headroom left by the single-pass baseline rather than with an independent effect of model release, tier, size, or weight availability. The comparisons are exploratory because each attribute is represented by limited pairwise contrasts and overlaps with other model characteristics.

4.2.4 Cross-Domain Evaluation

Purpose and setup

The cross-domain analysis focuses on the A/D contrast to examine whether the anchor-and-edit effect transfers beyond Biology. It uses the same dated Haiku main backbone (table 4.4) across the four retained BRIGHT splits, each evaluated in five replications; the three evaluation domains retain 90, 80 and 75 queries. The full A–D comparison is reported for the Biology design experiment (section 4.2.2). All four designs were run in each domain, and the complete-case sample is defined across A–D, so a formatting failure on any design drops that query for all of them; this section presents only the A/D contrast, because D is the proposed architecture and B and C are alternative coordination designs whose cross-domain behaviour is not the focus here.

Results

Table 4.6 and figure 4.3 report single-pass (A) and anchor-and-edit (D) mean Hit@1 for each domain. D shows its largest gain over A in Biology, smaller gains in Psychology and Sustainable living, and little or no gain in Earth science. This pattern suggests that the effect of bounded revision is domain-dependent. The paired Biology D–A difference

is +0.113 Hit@1, with a pointwise 95% interval of $[+0.029, +0.198]$; among the four domains only Biology's interval excludes zero, and none of the gains remains significant after multiplicity correction.

Table 4.6: Single-pass (A) and anchor-and-edit (D) mean Hit@1 per domain over five replications (Biology has 96 complete cases in replication 5). D−A is the paired complete-case estimate (figure 4.3) and may differ from the column difference by up to 0.001. D sig. counts the replications (out of five) reaching the unadjusted McNemar threshold ($p < 0.05$).

Domain	A	D	D−A	D sig.
Biology	0.566	**0.680**	**+0.113**	**5/5**
Earth science	0.433	0.436	+0.002	0/5
Psychology	0.627	0.667	+0.040	0/5
Sustainable living	0.463	0.512	+0.050	0/5

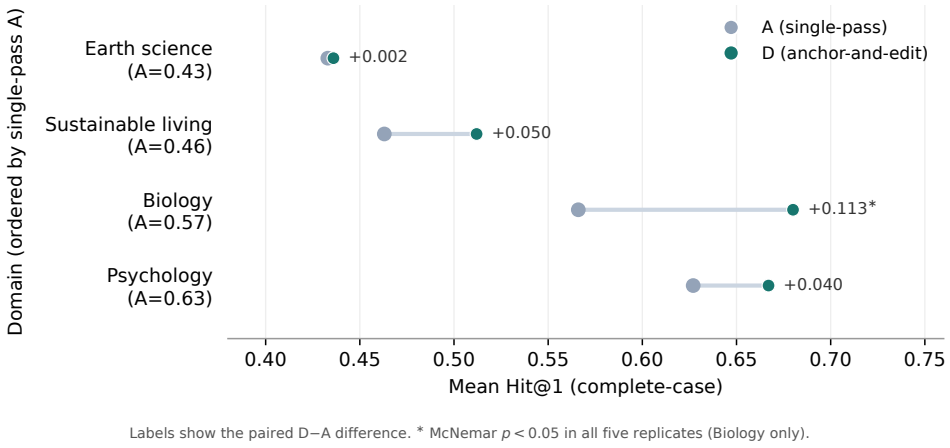


Figure 4.3: Domain-level single-pass (A) and anchor-and-edit (D) mean Hit@1, ordered by single-pass strength; labels show the paired D−A difference.

Domain-specific analysis

This post-hoc descriptive analysis examines why the D–A gain varies across domains. It uses the shared four-arm complete-case samples and aggregates query–run observations; repeated observations are not treated as independent samples. Table 4.7 reports, for each domain, how often D corrects a query that A ranked wrongly and how often it overturns a ranking that A had correct.

Biology combines the highest correction rate with relatively little disruption, whereas Earth science has the lowest correction rate. Its two percentages apply to different bases (corrected errors as a share of A's mistakes, overturned rankings as a share of A's correct results), so they cannot be compared directly; the underlying counts are

Table 4.7: Per-domain correction and disruption rates for D on the primary Haiku backbone (five pooled replications). The first rate is the share of query–run observations where A's top-1 was wrong and D moved a relevant document to rank one; the second is the share where A's top-1 was correct and D displaced it.

Domain	**A errors corrected by D**	**A correct rankings overturned by D**
Biology	36.7%	8.0%
Psychology	20.0%	5.5%
Sustainable living	21.9%	14.6%
Earth science	8.6%	10.8%

in fact similar, so the two kinds of change roughly offset. This is consistent with its near-zero net gain. Earth science query 34 illustrates the pattern: it asks whether an entire tectonic plate has ever subducted, and A ranks the gold document (which cites the Intermontane Plate) first, but D over-generalises the query and breaks the result in all five replications.

This example illustrates one failure mode in which concept-oriented reasoning overgeneralises a query requiring a specific instance or constraint. It does not by itself explain the domain effect, which may also reflect excerpt quality or Biology's role as the design domain.

4.2.5 Component Ablation

Purpose and setup

This ablation decomposes the anchor-and-edit design (D) to test whether its benefit can be attributed to any single component: the anchor (A's ranking, edited rather than rebuilt), the latent-concept reading, or the surface-oriented reading. Figure 4.4 shows the design and the component removed by each variant.

Table 4.8 presents a separate five-replication follow-up on the same BRIGHT Biology queries. Full D is compared directly with each component removal on that comparison's valid matched sample. All three contrasts retain 97 queries in every replication. Confidence intervals are pointwise and unadjusted; Holm correction applies only to the per-replication Hit@1 tests across the three contrasts.

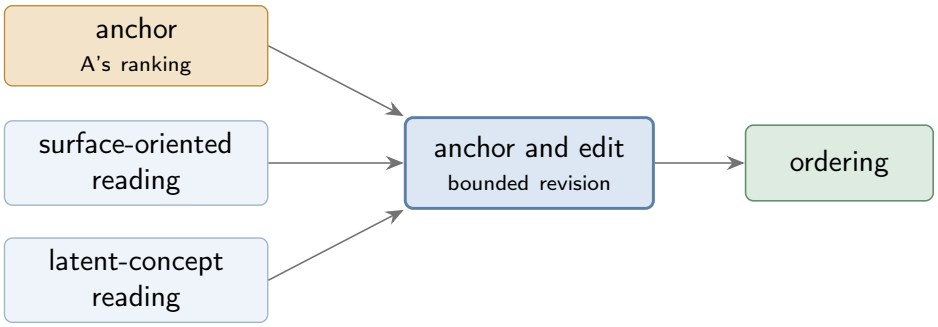


Figure 4.4: Anchor-and-edit (D) and its three components. Each ablated variant is full D with one component removed; table 4.8 reports the resulting change in effectiveness.

Results

Across five replications, mean Hit@1 is 0.573 for A, 0.643 for full D, 0.600 without the anchor, 0.590 without the latent-concept reading and 0.631 without the surface-oriented reading. The full D–A difference is 0.070, with a pointwise 95% query-cluster-bootstrap interval of $-0.017, +0.157$, which includes zero. This follow-up reuses the primary queries but has separate executions, so the results are reported separately and not pooled.

Table 4.8: Full-D minus ablation effects on Biology ($n = 97$, $k = 5$), with pointwise 95% bootstrap intervals.

Removed	**ΔHit@1**	**ΔMRR**
Anchor (A's ranking)	$+0.043$ $[-0.017, +0.107]$	$+0.044$ $[+0.003, +0.090]$
Latent-concept reading	$+0.054$ $[-0.025, +0.130]$	$+0.036$ $[-0.016, +0.088]$
Surface-oriented reading	$+0.012$ $[-0.025, +0.051]$	$+0.009$ $[-0.013, +0.033]$

The ablation does not establish that any individual component is necessary for D. All three Hit@1 intervals include zero, and no direct Hit@1 comparison reaches $p < 0.05$ in any replication, before or after Holm correction. The anchor-removal MRR interval excludes zero $(+0.003, +0.090)$, providing secondary, unadjusted evidence of a possible ranking benefit from the anchor; no multiplicity-adjusted MRR test is reported. Concept removal has the largest estimated Hit@1 effect, followed by anchor and surface removal, but this ordering is descriptive and does not demonstrate component importance or equivalence with full D.

4.2.6 Call-Matched Control

Purpose and setup

To test whether D's benefit comes from coordination rather than additional calls alone, we use A-SC@4, a call-matched control that repeats the single-pass baseline A four times with shuffled candidate orders and aggregates the rankings by mean rank, following the general principle of self-consistency and permutation-based ranking aggregation (Wang et al., 2023; Tang et al., 2024). It uses no concept or revision prompts, so both systems use four calls but only D applies anchor-and-edit reasoning. The control matches call count, not token budget, and represents one aggregation strategy. It is evaluated on the same 97 Biology queries over five paired replications.

Results

Table 4.9 combines A/D token counts from the metered run with those recorded for A-SC@4; effectiveness uses the paired follow-up.

Table 4.9: Call counts, token use and mean Hit@1 for A, A-SC@4 and D on Biology.

System	Calls/query	Tokens/query	Hit@1
A (single pass)	1	10,900	0.573
A-SC@4 (call-matched)	4	43,565	0.575
D (anchor-and-edit)	4	23,196	0.643

The call-matched control does not establish a statistically reliable advantage for D over repeated single-pass ranking. A-SC@4 is almost identical to A (0.575 vs. 0.573; $\Delta = +0.002$, 95% CI $-0.035, +0.043$). D exceeds A-SC@4 by $+0.068$, but the interval includes zero ($-0.017, +0.153$). The control matches call count rather than token budget, and A-SC@4 in fact uses more tokens than D (43,565 vs. 23,196 per query) while performing no better; the budgets are therefore not equalised, and the interval spanning zero means the comparison does not resolve whether D outperforms repeated single-pass ranking.

4.3 Study 2: Applied Ranking

Purpose and setup

Study 2 applies A–D to the 36 proposal–call pairs described in section 3.2.2, using the expert labels as the primary reference. Agreement is measured with three-level quadratic-weighted κ. The author's independent annotation provides the non-expert rubric reference (quadratic-weighted $\kappa = 0.49$), not a second gold standard. Given the small sample and single expert, all results are descriptive. A staged pairwise matcher is examined separately as an exploratory probe in appendix A.4.

Table 4.10: Expert relevance grades by proposal. Each proposal has six funding calls. Proposal 6 is labelled `no_match` on every call and is excluded from all ranking metrics. Proposal 5 has no `match` label but one `borderline` call, so it contributes to graded nDCG@6 (borderline gain = 1) but is excluded from the match-only Hit@1 and MRR.

Proposal	`match`	`borderline`	`no_match`
Proposal 1	1	2	3
Proposal 2	2	1	3
Proposal 3	3	0	3
Proposal 4	2	2	2
Proposal 5	0	1	5
Proposal 6	0	0	6
Total	8	6	22

Table 4.11 reports the applied listwise evaluation on the main backbone. Variant D uses a promotion cap of one because each proposal has six candidate calls. Each proposal and funding call is represented by its full document text, and each variant is run once with a fixed presentation-order seed (42). Agreement uses all 36 proposal–call pairs, while the ranking metrics use the eligibility rules defined in section 3.5. The table therefore reports a single-seed evaluation (seed 42); presentation-order sensitivity is not characterised for the applied task, a limitation accepted because Study 2 is a small descriptive feasibility study and its full-text runs are comparatively costly.

Results

On this small, near-ceiling set, all variants rank a relevant call first for the same three of four match-eligible proposals, and nDCG@6 ranges only from 0.949 to 0.954. D inherits A's grades by design, so its agreement (κ) equals A's regardless of any edits; because D also makes no promotion on any proposal here, its ranking, and hence nDCG@6, Hit@1

Table 4.11: Applied listwise results on the main backbone (single seed).

Variant	κ	nDCG@6	Hit@1	MRR
A	**0.62**	0.949	0.75	0.83
B	0.58	0.951	0.75	**0.88**
C	0.56	**0.954**	0.75	**0.88**
D	**0.62**	0.949	0.75	0.83

and MRR, is identical to A's as well. D's effect is therefore confined to the ranking columns, where on this set it makes no difference.

All variants exceed the non-expert rubric reference in agreement with the expert ($\kappa = 0.56$–0.62 versus 0.49), but this does not establish expert-level assessment. The reference is not an expert ceiling, and the bootstrap difference for A includes zero ($+0.12$ on unrounded values, 95% CI $[-0.14, +0.41]$). With only four or five eligible proposals, the data cannot distinguish limited headroom, small sample size and limited benefit from coordination.

Cross-model panel

A single backbone cannot show whether this null result is specific to `haiku-4.5` or general. The applied task was therefore repeated on nine backbones under the same protocol (single seed, full document text), spanning proprietary and open-weight models. Table 4.12 reports system–expert agreement and nDCG@6 ranking quality for the single-pass baseline (A) and anchor-and-edit (D) on each backbone.

Table 4.12: Applied listwise results across nine backbones. All runs use one presentation-order seed and full document text. κ is agreement with expert grades; D inherits A's grades, so its effect appears only in the ranking columns. The main backbone is highlighted.

Backbone	κ	nDCG@6 (A)	nDCG@6 (D)	D−A
`haiku-4.5`	0.62	0.95	0.95	0.00
`opus-4-6`	0.71	0.97	0.99	+0.02
`sonnet-4-6`	0.72	0.97	0.97	0.00
`gemini-3.5-flash`	0.67	1.00	0.90	−0.10
`gemini-3.7-flash`	0.66	0.95	0.95	0.00
`Llama-3.3-70B`	0.21	0.80	0.78	−0.02
`Qwen3.5-9B`	0.55	0.95	0.95	0.00
`DeepSeek-V4-Flash`	0.73	0.98	0.98	0.00
`gpt-oss-120b`	0.72	0.97	0.97	0.00

The panel shows substantial variation in agreement across backbones, from $\kappa = 0.21$ for `Llama-3.3-70B` to 0.73 for `DeepSeek-V4-Flash`. Eight of nine backbones meet or exceed the non-expert rubric reference of 0.49. This variation in agreement does not translate into a consistent effect of anchor-and-edit: D leaves nDCG@6 essentially unchanged on eight of the nine backbones (differences within ±0.02) and reduces it only on `gemini-3.5-flash` (1.00 → 0.90). The applied task therefore shows model-dependent agreement but little consistent headroom for bounded revision.

The small candidate sets and high single-pass scores leave limited room for bounded revision to improve the leading recommendation. Although this resembles the benchmark pattern of larger gains from weaker baselines, the descriptive panel cannot distinguish ceiling effects, task structure and model-specific behaviour.

4.4 Anchor-Visibility Perturbation Check

Purpose and setup

Study 1 hides A's ranking from the coordinator, whereas Study 2 shows it as an anchor to preserve or edit. This controlled perturbation check asks whether showing the anchor induces sycophancy (Perez et al., 2022; Sharma et al., 2023), which is mechanical preservation of the ranking the coordinator is shown rather than independent revision.

The check uses the 53 of 97 Biology queries where A ranks a relevant document first. For each, a degraded ranking moves every relevant document below the non-relevant documents, creating an error recoverable by one promotion. With the queries, model, seed, prompt and one-promotion cap fixed, the hidden coordinator names a promotion without seeing the ranking; the visible coordinator sees the ranking with a preserve-unless-warranted default. Each is applied to A's correct ranking and to the degraded ranking.

Table 4.13: Anchor-visibility perturbation check on Biology ($n = 53$), promotion cap one, single seed. Rows differ in anchor presentation.

Configuration	**Recovery (degraded)**	**Preservation (correct)**
Hidden (used in Study 1)	87% (46/53)	94% (50/53)
Visible (used in Study 2)	94% (50/53)	81% (43/53)

Results

Recovery is high in both conditions (87% hidden; 94% visible; paired McNemar $p = 0.22$), so the visible coordinator does not mechanically preserve a degraded anchor. On correct rankings, however, visible preservation is lower (81% versus 94%; three versus ten breaks; $p = 0.04$). The visible coordinator is therefore no more conservative than the hidden one, so sycophancy is not observed in this check. This weakens simple anchor deference as an explanation for D's zero edits in Study 2. However, as the check uses constructed errors from Study 1, it cannot rule out deference on the applied documents.

4.5 Computational Cost and Outcome Stability

4.5.1 Computational Cost

Purpose and setup

This analysis measures the expenditure associated with additional calls. For BRIGHT Biology, costs are taken from a 97-query metered ablation run. D's incremental cost covers its two hypothesis calls and bounded revision; its end-to-end cost also includes the A ranking used as the anchor. These are reported separately. Applied costs are taken from the probe containing six proposals and six candidate calls per proposal. Cost coverage is incomplete: the retained audit does not contain BRIGHT B/C metering, so no complete A–D cost comparison is claimed for that dataset.

Results

Table 4.14 reports BRIGHT Biology cost from a metered run. Costs use the pinned Haiku pricing of $1.00 per million input tokens and $5.00 per million output tokens.

Table 4.14: BRIGHT Biology costs from a metered run ($n = 97$); per-query values are approximate.

Variant	$/query	$ total	Calls
A	0.0133	1.29	97
D incremental	0.0158	1.53	291
A+D end-to-end	0.0291	2.82	388

D's end-to-end cost is approximately 2.19 times A's, including the baseline ranking and three additional calls per query. The incremental figure assumes that A's ranking is

already available.

On the applied probe of six proposals with six candidate calls each, the recorded costs are \$0.056 for A, \$0.071 for B, \$0.059 for C, and \$0.088 for D. These are the original probe totals; the BRIGHT audit does not establish the same stage accounting for the applied probe, so an applied end-to-end cost ratio is not inferred from them.

4.5.2 Outcome Stability

Purpose and setup

This analysis examines whether additional coordination changes outcome consistency across saved replications. A query is stable when its top-1 correctness is unchanged across replications.

Stability is reported descriptively for the controlled benchmark, as the number of queries whose top-1 correctness is unchanged across the five replications.

Results

Table 4.15: Top-1 correctness stability across five Biology replications ($n = 97$).

Variant	Stable queries	Queries changing correctness
A	97/97 (100%)	0
D	91/97 (93.8%)	6

As shown in table 4.15, A's correctness indicator is unchanged for every query, while D's changes for six queries. This measures correctness stability, not identity of the selected document or of the full ranking. First-gold ranks vary across saved A outputs. As detailed in section 4.1.2, incomplete request provenance prevents ruling out partial cache reuse, so these figures describe saved-output agreement.

Discussion

This chapter interprets the findings against the research questions and hypotheses. Overall, the evidence supports bounded revision as a task-dependent ranking strategy: it can improve a baseline when relevance remains difficult to resolve, but its benefit is not uniform across domains, backbones or applied settings.

5.1 Ranking effectiveness and component contributions

The evidence provides partial support for H1: D improves A most clearly on Biology, but not uniformly across domains or backbones. In the captured Biology replication, ten of fifteen corrections promote documents from ranks 2–5 (section 4.2.2). The anchor-visibility check further shows that a visible coordinator recovers 50 of 53 injected errors, rather than mechanically preserving its anchor, but it preserves correct rank ones less often than the hidden condition (81% versus 94%; section 4.4). This mechanism evidence is limited to large injected errors on the design domain.

The ablations do not identify a necessary component: every component-removal Hit@1 interval includes zero (table 4.8). Larger gains also coincide with weaker A baselines, as predicted by H2, but baseline strength is part of the gain calculation and is confounded with model attributes. The result is therefore a descriptive association, not a validated mechanism or a rule for selecting when to apply D.

5.2 Applied ranking

Study 2 addresses RQ2 by examining agreement with expert grades and changes in candidate ordering. On the main backbone, the listwise systems achieve quadratic-weighted κ values of 0.56–0.62, compared with 0.49 for the non-expert rubric reference (section 4.3). The interval for the single-pass system's agreement difference from this reference includes zero, so this comparison does not establish that it outperforms the second annotator. Model and expert judgements remain imperfectly aligned despite receiving the same written criteria.

The additional value of coordination is less evident. D inherits A's grades, so its contribution must be assessed through ranking changes rather than agreement. On the main backbone, A and D have identical aggregate ranking scores. Across the applied model panel, D leaves the ranking essentially unchanged in most settings and slightly worsens it for one backbone (table 4.12). These observations provide no consistent evidence that bounded revision improves the applied ranking. Because the single-pass rankings are already close to the best achievable ordering, there is little room for revision to help, but the small dataset cannot establish whether this limited headroom is the reason. One alternative explanation is that showing D the anchor made it conservative, so its inaction on the applied set reflects visibility rather than an absence of recoverable errors; but the anchor-visibility check (section 4.4) weakens this, because a visible coordinator did not become more conservative than a hidden one on Biology. It does not show that the applied rankings lacked recoverable errors, because the check uses constructed Biology errors rather than applied documents.

5.3 Cost–benefit trade-off

The findings support bounded revision as a practical option for systems that can justify a few additional model calls per query. On BRIGHT Biology, end-to-end anchor-and-edit costs about 2.19 times the single-pass baseline (section 4.5.1), and the call-matched control shows that D's point estimate exceeds repeated single-pass ranking while the direct comparison remains statistically unresolved (section 4.2.6). Deployments should therefore treat bounded revision as a favourable but unconfirmed trade rather than an established win, weighing its extra cost against the value of correcting the leading recommendation.

5.4 Ethical considerations

This study raises ethical considerations concerning the governance of proprietary data and the use of automated recommendations in resource-allocation decisions. The applied dataset was accessed under authorisation from the UCL Translational Research Office. Proposal and funding-call texts, expert grades and derived artefacts remain internal; no identifying details, data or labels are released. Because the proposals are unpublished, sending their texts to third-party inference APIs carries a residual disclosure risk beyond the providers' own data-handling policies.

In any deployment, the original and revised rankings should be retained, promoted

candidates should be recorded, and human review should be available for consequential changes. The non-public data should be used only under authorised access, and prompts, source texts and saved responses handled carefully, particularly when external model providers are involved.

Listwise LLM ranking is sensitive to order and position (Liu et al., 2024; Zheng et al., 2023). While presentation-order seeds reduce this confound and the perturbation check separately tests anchor visibility (section 4.4), position bias is not measured directly. In addition, automated matching may favour language or topics overrepresented in model training, reproducing existing allocation biases (Mehrabi et al., 2021). Finally, the single-practitioner labels are subjective (Voorhees, 2000) and are treated as a comparison point, not ground truth.

Proposal–funding matching influences the allocation of limited resources, so an incorrect recommendation could unfairly disadvantage a proposal. Given these considerations, the system is intended for decision support with human review rather than autonomous replacement of expert judgement. The observed agreement does not demonstrate expert-level assessment.

5.5 Limitations

The evidence has several limits. First, the primary benchmark gains do not survive multiplicity correction, and broader domain and backbone coverage would strengthen them. Second, the coordination prompts, including the concept-abstraction step, were developed on Biology and use science-oriented language (`mechanism, principle, compound, or law`; see appendix A.2). They were deliberately applied to the other domains without adaptation, so that cross-domain differences could be attributed to the coordination strategy rather than to per-domain prompt tuning. A consequence of this controlled-transfer choice is that the weaker effect outside Biology may reflect prompt specialisation as much as a genuine absence of headroom, and whether domain-adapted prompting would recover it is an open question. Third, the applied study is small and descriptive: it uses six proposals, one expert reference, metric-specific ranking subsets and complete documents supplied without field extraction or chunking, and it does not test rejection when no acceptable candidate exists. Fourth, the second annotation was produced by the author rather than an independent domain expert, under the same rubric supplied to the systems, so the Study 2 agreement comparison cannot separate rubric-following from domain knowledge, and κ estimated from only 36 pairs and a single expert is only indicative. Fifth, the anchor is hidden from the coordinator in Study 1 but visible in Study 2, so the two studies evaluate dataset-specific implementations of

the same design rather than an identical system. The perturbation check in section 4.4 indicates that error recovery is comparable under both configurations, but the applied results are not a strict replication of the benchmark configuration.

Future Work

The findings suggest three practical directions for extending this work.

First, bounded revision should be evaluated across more domains and model backbones. The strongest evidence here comes from Biology, where the design was developed, while effects elsewhere are less consistent. Larger, baseline-matched panels with complete output records would help determine when revision transfers and whether gains reflect the quality of the initial ranking or properties of a particular model. Replications should keep the candidate pool and relevance objective fixed so that changes can be attributed to the coordination strategy.

Second, future work should make revision more selective. The ablations do not identify a necessary component, but they leave open how the components interact and when an edit is warranted. A system could retain the initial ranking when additional evidence is weak and revise only when the expected correction outweighs the risk of a harmful promotion. This decision could use simple signals from the competing readings or from disagreement between stages. Evaluation should consider useful corrections, introduced errors and the cost of extra calls together, rather than treating a higher ranking score as the only objective.

Third, the applied study should be repeated on larger and more varied proposal–funding datasets with multiple independent experts. This would provide a stronger reference for agreement and help separate annotator variation from system variation. The data should include proposals for which no suitable call exists, since a useful matcher must support rejection as well as ranking. Although outside the present scope, future matching systems should first extract and expose decision-relevant fields, such as eligibility criteria, deadlines and scientific fit, from proposals and funding calls before passing them to the agentic coordination pipeline. They should also preserve the original and revised recommendations and keep human review in the loop, so that potentially suitable opportunities are not removed without oversight.

Conclusion

This dissertation investigated whether coordinating multiple language-model calls can improve relevance assessment when the available documents remain fixed. The motivation was an applied problem that has received limited attention: matching research proposals with funding opportunities requires judgements about scientific fit, eligibility and context, yet these judgements are difficult to reduce to lexical overlap or a single model call. The dissertation therefore combined a controlled benchmark study of document ranking with an applied study of proposal–funding call matching. Together, the studies examined ranking quality, agreement with expert judgements, the contribution of individual components and the practical cost of additional model calls. The research questions and hypotheses are discussed in the Discussion chapter.

The central methodological contribution is anchor-and-edit, a bounded-revision architecture for listwise ranking. It retains an initial ranking as an anchor and uses additional reasoning to make only a small number of targeted promotions. This design preserves useful ordering information while allowing specific ranking errors to be revisited. The evaluation compared it with a single-pass baseline and two alternative coordination designs under a shared candidate pool, relevance objective and model configuration. Component ablations, a call-matched control, cross-domain comparisons and cross-model analyses were included to distinguish the effect of the design from the effect of additional computation alone.

All four LLM designs outperform the tested lexical and cross-encoder baselines on the Biology candidate pools, while the additional gains from coordination vary across designs and settings. Study 1 shows that bounded revision can improve document ranking in several settings, with the clearest and most consistent evidence in Biology, where the design was developed. The gains are less uniform across domains and backbones, and the component tests do not identify a single necessary part of the architecture. The case analysis illustrates how rankings change under revision: the system can recover relevant documents that were initially close to the top, or in some cases ranked far lower, but it can also disrupt a correct ranking. One possible explanation for the uneven gains is that bounded revision helps most when the initial ranking leaves recoverable errors near the top, though this remains to be validated.

Study 2 is a descriptive external test of proposal–funding matching. The listwise systems agree with the expert only about as closely as a non-expert applying the same rubric,

so they do not demonstrate expert-level assessment. Anchor-and-edit shows no clear advantage here and made no edits, because the candidate set is small and the baseline often already strong, leaving little headroom to correct. This is consistent with Study 1, where anchor-and-edit gave little benefit when the single-pass baseline was already strong.

These contributions come with clear limitations. The benchmark gains do not survive multiplicity correction, the coordination prompts were developed on a single domain, and the applied study is small and descriptive, resting on six proposals and a single expert reference. Because the applied data is proprietary, its use also raises data-governance and disclosure obligations, and any deployment that influences resource allocation must retain human oversight rather than replace expert judgement. Future work is therefore proposed to broaden domain and backbone coverage, to make revision more selective about when an edit is warranted, and to repeat the applied evaluation on larger and more varied proposal–funding datasets with multiple independent experts (chapter 6).

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Appendix

A.1 Reported Backbone Versions and Release Dates

Table A.1 gives the public release date of each backbone in the reported cross-model panel (table 4.4). Where a model identifier pins a dated snapshot, the snapshot date is listed; otherwise the date is the public release of the version used. Dates were taken from the providers' official announcements and model documentation.

Table A.1: Release dates of the ten reported cross-model backbones.

Model	**Release date**
`meta-llama/Llama-3.3-70B-Instruct-Turbo`	2024-12-06
`openai/gpt-oss-20b`	2025-08-05
`openai/gpt-oss-120b`	2025-08-05
`claude-haiku-4-5-20251001`	2025-10-01
`claude-opus-4-6`	2026-02-05
`claude-sonnet-4-6`	2026-02-17
`Qwen/Qwen3.5-9B`	2026-03-02
`gemini-3.5-flash`	2026-05-19
`deepseek-ai/DeepSeek-V4-Flash-0731`	2026-07-31
`gemini-3.7-flash`	2026-08-13

A.2 Study 1 Prompts (BRIGHT reranking)

All four variants share the same candidate set, candidate rendering, and relevance objective (defined by the A-prompt): A and B rank with the A-prompt directly (B conditions it on an abstract concept), while the competing-hypotheses (C) and anchor-and-edit (D) variants add a coordination step with its own prompt and do not re-tune the relevance objective. Angle brackets `<...>` mark interpolated inputs (the question, the rendered candidate documents, and intermediate hypotheses); the full runners are in the accompanying code. Every prompt requests JSON only, decoded at temperature zero.

Single-pass baseline (A).

```
You are given a question and 100 candidate documents (opaque ids). Rank
ALL ids
from most to least relevant for answering the question. Return JSON only.

Schema: {"ranked_ids": ["<id>", ...]}

Question: <question>

Documents:
<candidate documents: [{"id": ..., "text": ...}, ...]>
```

Concept-abstraction agent (B).

```
You are a scientific concept-abstraction agent. Read the question and
identify
the LATENT scientific concept(s) that actually answer it -- the underlying
mechanism, principle, compound, or law -- NOT merely the surface topic the
question mentions. Give concise reasoning. If genuinely uncertain, give one
lower-confidence alternative concept. Return JSON only.
Schema: {"concept":"<the core latent concept(s)>",
         "reasoning":"<1-2 sentences: why this concept, not the surface
         topic>",
         "alternative":"<optional alternative concept, or empty string>"}
Question: <question>
```

Concept-conditioned rerank (B).

```
You are given a question and 100 candidate documents (opaque ids). Rank
ALL ids
from most to least relevant for answering the question. Return JSON only.

Schema: {"ranked_ids": ["<id>", ...]}

Latent concept hypothesis (from the concept-abstraction step; treat as a
strong
hint about what the question is really about, but rank by the documents):
  concept: <concept>
  reasoning: <reasoning>
  alternative: <alternative>

Question: <question>
```

```
Documents:
<candidate documents: [{"id": ..., "text": ...}, ...]>
```

Surface/lexical hypothesis agent.

```
You are the SURFACE/LEXICAL hypothesis agent. Describe what the question
LITERALLY asks -- the surface reading, terms an answer doc would contain,
answer
type, and any hard constraints. Do NOT abstract to deeper principles.
Return JSON only.
Schema: {"surface_intent":"<one sentence>","key_terms":["..."],
         "expected_answer_type":"<...>","explicit_constraints":["..."]}
Question: <question>
```

Latent-concept hypothesis agent.

```
You are the LATENT CONCEPT hypothesis agent. Identify the underlying
concept(s)
that actually answer the question -- mechanism, principle, compound, or
law -- NOT
the surface topic. Alternatives are first-class. Return JSON only.
Schema: {"primary_concept":"<...>","reasoning":"<1-2 sentences>",
         "concept_terms":["..."],
         "alternatives":[{"concept":"<...>","confidence":"high|medium|low
         "}]}
Question: <question>
```

Anchor-and-edit coordinator (D).

```
You are the COORDINATOR in an ANCHOR-AND-EDIT retrieval system. A
single-pass
baseline has ALREADY produced a good ranking of the candidates (you do not
see
it). Your job is NOT to re-rank everything. Your job is to decide whether
the
baseline's surface reading suffices, and if not, to name the FEW documents
that
the correct latent reading promotes to the top.

<surface-oriented and latent-concept readings>
```

```
Decide:
- decision="surface_sufficient": the answer is the document a
literal/lexical
  reading would pick; the baseline is trusted UNCHANGED. promote_ids MUST
  be [].
- decision="concept_decisive": the answer is a conceptually-distant
document that
  only the latent concept (or one of its alternatives) explains. Put in
  promote_ids the 1-2 candidate ids that best embody that concept, best
  first.
- decision="mixed": ambiguous; promote_ids may hold 1-2 ids you are
confident
  belong at top.
Reason from the EVIDENCE in the candidate texts. Do NOT list more than 2
promote_ids. Return JSON only.
Schema: {"decision":"surface_sufficient|concept_decisive|mixed",
         "promote_ids":["<id>"],"rationale":"<short, evidence-based>"}
Question: <question>

Documents:
<candidate documents>
```

Full-rerank coordinator (competing hypotheses, C; also the no-anchor ablation).

```
You are the COORDINATOR: a reasoning arbiter, NOT a rule engine and NOT a
score-averager. You are given the question, TWO competing hypotheses about
what
it is really asking, and the actual candidate documents. Decide -- using
the
EVIDENCE in the documents -- which reading the answer supports, and
produce the
final ranking.

Two competing hypotheses (neither is privileged):
  SURFACE/LEXICAL: <surface-oriented reading>
  LATENT CONCEPT : <latent-concept reading>

Rules:
- If the documents show the surface reading already identifies the right
answer,
```

```
  rank by it.
- If the answer is a conceptually-distant document that only the latent
concept
  explains, rank that to the top.
- Do NOT average. Reason document-by-document for the top of the ranking.
Rank
  ALL 100 ids.
Return JSON only.
Schema: {"ranked_ids": ["<all ids, most->least relevant>"]}
Question: <question>

Documents:
<candidate documents>
```

A.3 Study 2 Prompts and Annotation Rubric

The annotation rubric below was given to the human annotators and is inserted verbatim into the matching prompts, so grader and model apply identical criteria (section 4.3). Where a prompt injects it, the point is marked `[ANNOTATION RUBRIC]`.

Annotation rubric (expert reference and model criteria).

```
GRADING CRITERIA (apply exactly; grade using ONLY the proposal and
funding-call
texts below, not outside knowledge of the sponsor or its strategy):
- "match": the proposal clearly falls within the call's stated therapeutic
areas
  / scope / eligibility.
- "borderline": partial fit -- overlaps on some dimension (disease area,
modality,
  mechanism) but has a scope, eligibility, or sponsor-specific mismatch
  that makes
  suitability genuinely uncertain.
- "no_match": outside the call's stated areas/scope, or violates a stated
  constraint (wrong disease area, wrong modality, an explicit exclusion,
  or a
  sponsor/institutional conflict the call rules out).
Decision procedure, in order: (1) read the call's stated areas of interest
/
```

```
scope / eligibility; (2) read the proposal's disease area,
mechanism/modality,
stage, and any constraints; (3) if a hard constraint is violated, grade
"no_match"; (4) else if it squarely fits, grade "match"; (5) otherwise
grade
"borderline".
```

Listwise grading (A single-pass; B adds a concept hint).

```
You are matching a research PROPOSAL to candidate FUNDING CALLS / company
wishlists. Rank ALL 6 candidate ids from best to worst fit for this
proposal,
and assign each a relevance grade.

[ANNOTATION RUBRIC]

Return JSON only.
Schema: {"ranked_call_ids": [all ids best-to-worst],
         "grades": {"<id>": "match"|"borderline"|"no_match", ... for every
         id},
         "rationale": "<=2 sentences"}
<for variant B only: proposal research-intent concept hint>
PROPOSAL:
<proposal text>

CANDIDATE CALLS:
<six funding calls: [{"id": ..., "text": ...}, ...]>
```

Competing-hypotheses coordinator (C).

```
You are the COORDINATOR in a COMPETING-HYPOTHESES matcher. You are given
the
proposal, TWO competing readings of what it is really about (neither
privileged), and the candidate calls. Decide from the evidence in the call
texts
which reading the best matches support, and produce the FINAL ranking of
ALL
calls from best to worst fit, assigning each a relevance grade. Reason
call-by-call; do NOT average.

[ANNOTATION RUBRIC]
```

```
Return JSON only.
Schema: {"ranked_call_ids": [all ids best-to-worst],
         "grades": {"<id>": "match"|"borderline"|"no_match", ...},
         "rationale": "<=2 sentences"}
SURFACE reading: <surface-oriented reading>
LATENT reading: <latent-concept reading>
PROPOSAL: <proposal text>
CANDIDATE CALLS: <six funding calls>
```

Anchor-and-edit coordinator (D). Grades are inherited from A; this stage only decides promotions.

```
You are the COORDINATOR in an ANCHOR-AND-EDIT reranker. A single-pass
baseline
has ALREADY ranked the candidate calls (the ANCHOR, best first). Your
DEFAULT is
to PRESERVE it. Only if the surface/latent interpretations give STRONG
evidence
that specific calls belong at the very top should you name those ids (best
first). Do NOT re-rank from scratch; do NOT edit on weak or stylistic
grounds.

ANCHOR ranking (best first): <A's ranking>
SURFACE interpretation: <surface-oriented reading>
LATENT concept: <latent-concept reading>

Return JSON only.
Schema: {"action":"keep"|"edit","promote_ids":["<id>", ... best first, or
empty],
         "rationale":"<=2 sentences"}
PROPOSAL: <proposal text>
CANDIDATE CALLS: <six funding calls>
```

Staged matcher: proposal understanding.

```
You are the PROPOSAL UNDERSTANDING agent. Read ONLY the proposal. Do NOT
consider
any funding call. Produce a structured semantic representation. Return
JSON only.
Schema: {"research_problem":"...","disease_area":"...",
```

```
         "scientific_question":"...","mechanism":"...",
         "technology_or_modality":"...","research_stage":"...",
         "expected_impact":"...","keywords":["..."]}
PROPOSAL: <proposal text>
```

Staged matcher: funding-call understanding.

```
You are the FUNDING CALL UNDERSTANDING agent. Read ONLY the funding call.
Do NOT
read any proposal. Extract what kind of projects the programme is seeking.
Return
JSON only.
Schema: {"programme_goal":"...","target_disease":"...",
         "preferred_mechanisms":["..."],"preferred_modalities":["..."],
         "technology_scope":"...","eligibility_constraints":["..."],
         "desired_research_stage":"...","success_criteria":["..."]}
FUNDING CALL: <funding call text>
```

Staged matcher: alignment (reads the two structured summaries).

```
You are the ALIGNMENT agent. You receive ONLY two structured summaries. Do
NOT
reread the original documents. Evaluate semantic alignment
dimension-by-dimension
(0-100 each). Return JSON only. Do NOT produce a final decision.
Schema:
{"thematic_fit":0-100,"mechanistic_fit":0-100,"technology_fit":0-100,
         "stage_fit":0-100,"constraint_fit":0-100,"major_strengths":["...
         "],
         "major_mismatches":["..."],"uncertainties":["..."]}
<two structured summaries>
```

Staged matcher: final judge.

```
[ANNOTATION RUBRIC]
You are the FINAL DECISION agent for proposal--funding-call matching.
Using all
inputs below, produce the final judgement. Return JSON only.
Schema: {"match_score":0-100,"decision":"match"|"no_match"|"borderline",
         "confidence":0-100,"reasoning":"<=3 sentences",
         "key_supporting_dimensions":["..."]}
PROPOSAL: <proposal text>
```

```
FUNDING CALL: <funding call text>
PROPOSAL SUMMARY: <proposal summary>
FUNDING CALL SUMMARY: <funding call summary>
ALIGNMENT ANALYSIS: <alignment dimensions>
```

A.4 Staged Pairwise Matcher (Exploratory)

As a secondary probe, a decomposed *staged* matcher grades individual proposal–call pairs instead of ranking the six calls as the A–D listwise designs do. It takes one proposal and one funding call and returns a single grade (`match`, `borderline`, or `no_match`) over the same 36 pairs and expert labels used in the applied evaluation (section 4.3). Because it changes the judgement form (per-pair grading) as well as the architecture, it is not directly comparable to the listwise designs and is reported here as an exploratory observation rather than an architectural comparison. It is descriptive ($n = 36$, single expert).

Four stages run in sequence (figure A.1). The first reads the proposal alone and produces a structured summary; the second does the same for the call; the third receives the two summaries and produces an alignment assessment; the fourth receives the raw proposal, raw call, both summaries, and the alignment, then issues the grade under the shared rubric (section 3.2.2). The design separates proposal understanding, call understanding, alignment and final judgement so that intermediate representations can be reused and inspected, following the broader principle of decomposing complex reasoning across stages (Zhou et al., 2023; Hong et al., 2024). The first two stages are computed once per document and reused across pairs. Prompts are listed in appendix A.3.

The staged matcher reaches quadratic-weighted $\kappa = 0.27 \pm 0.01$ (mean $\pm$ SD across ten replications) against the expert grades, below the non-expert rubric reference of 0.49 and the listwise designs' range of 0.56–0.62 (section 4.3). The matcher is given the expert's rubric verbatim, so the shortfall arises under a shared rubric rather than from withholding the criteria; it does not, however, establish that the criteria are unambiguous, since a shared rubric can still be interpreted differently. Its errors are concentrated on the positive class: under the rubric it never assigns `match`, resolving every pair to `no_match` or `borderline`. In a representative replication it returns `no_match` on all 22 expert `no_match` pairs with no false positives, but recovers none of the eight expert `match` pairs (grading five `no_match` and three `borderline`). Following the rubric's hard-constraint-first procedure, it finds a disqualifying mismatch in almost every positive pair. Its decisions are stable, unchanged on 35 of 36 pairs across the ten runs. With only 36 pairs and a single expert, these are descriptive observations

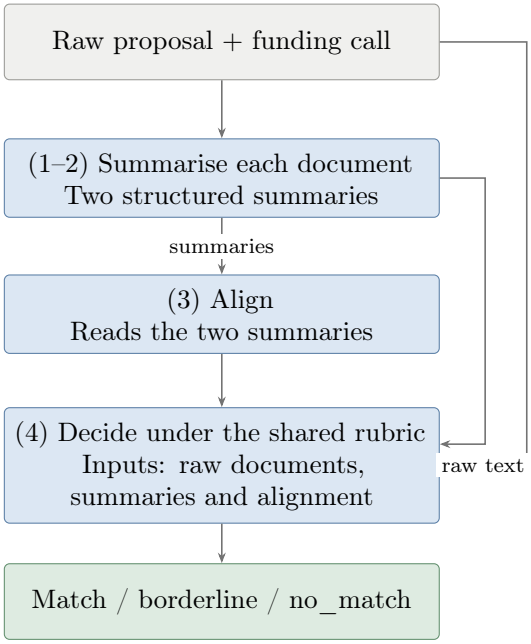


Figure A.1: Staged pairwise matcher (summary-fed configuration). Four stages run in sequence: each document is summarised (1–2), the two summaries are aligned (3), and the final stage (4) issues the grade from the raw documents, summaries and alignment. Stages 1–2 are computed once per document and reused across pairs.

that motivate human review before excluding opportunities, not a general pattern. The failure mode is uniform: following the rubric's hard-constraint-first procedure, the matcher treats a stated call preference (such as a preference for clinical-stage assets) as a disqualifying constraint and so under-calls proposals that the expert grades as strong scientific fits, while it hedges correctly on genuinely uncertain pairs. To protect the non-public proposal–funding matching data, individual proposal–call examples are not reproduced here.